

Experiment No. 08

Random Forest and Boosting Comparison

Aim:

To implement and compare **Decision Tree, Random Forest, AdaBoost, and Gradient Boosting regression models**, and analyze their performance, stability, training time, and feature importance using cross-validation.

Objective:

After completing this experiment, students will be able to:

- To understand the concept of **ensemble learning**.
- To understand **bagging** and **boosting** techniques.
- To implement a Decision Tree Regressor as a baseline model.
- To implement a Random Forest Regressor using multiple decision trees.
- To implement AdaBoost and Gradient Boosting regressors.
- To compare the models using **cross-validation**.
- To measure the training time of different ensemble methods.
- To analyze the stability of models using cross-validation standard deviation.
- To investigate feature importance in Random Forest and Boosting models.
- To recommend the most suitable model based on performance, stability, training time, and interpretability.

Pseudo Algorithm :

Steps:

1. **Start**
2. Load the house-price dataset.
3. Inspect the dataset and identify features and target.
4. Handle missing values and unnecessary columns.
5. Separate the features XX and target yy.
6. Split the data into training and testing sets.
7. Create a Decision Tree Regressor.
8. Create a Random Forest Regressor.
9. Create an AdaBoost Regressor.
10. Create a Gradient Boosting Regressor.
11. Train each model using the training dataset.
12. Perform **5-fold cross-validation** on the training data.
13. Calculate the cross-validation mean and standard deviation.
14. Measure the training time of each model.
15. Compare the models based on RMSE/MSE or R^2 .
16. Train Random Forest on the complete training data.
17. Extract and sort Random Forest feature importances.
18. Extract feature importance from the boosting model.
19. Compare model performance, stability, training time, and interpretability.
20. Select the most suitable model.
21. **Stop**

Sample Outcome:

- The Decision Tree, Random Forest, AdaBoost, and Gradient Boosting regression models were successfully implemented.
- The models were evaluated using 5-fold cross-validation, and their mean RMSE, standard deviation, and training time were recorded.
- The ensemble methods generally provided more stable predictions than a single decision tree. Random Forest reduced the variance associated with an individual decision tree through bootstrap sampling and feature randomisation, while boosting methods improved predictions by sequentially learning from previous errors.
- Feature importance was also extracted from Random Forest and Gradient Boosting to identify the most influential house characteristics.

Model	CV RMSE Mean	CV RMSE Std	Training Time
Decision Tree	165,000	12,000	0.05 s
Random Forest	105,000	5,500	2.80 s
AdaBoost	135,000	7,000	0.80 s
Gradient Boosting	100,000	4,800	1.50 s

Sample Observation:

- The Decision Tree provides a simple and interpretable baseline.
- Random Forest generally provides more stable predictions because it combines multiple trees.
- AdaBoost improves upon weak learners by focusing on observations that were previously predicted poorly.
- Gradient Boosting can achieve strong predictive performance by sequentially correcting errors.
- Ensemble models generally require more training time than a single decision tree.
- A lower CV RMSE indicates better regression performance.
- A lower CV standard deviation indicates greater stability across folds.
- Feature importance from Random Forest can be used to identify the most influential predictors.
- The final model should not be selected using accuracy alone; performance, stability, computational cost, and interpretability should all be considered.

Tools / Software Required:

Hardware: Computer/Laptop, Minimum 4 GB RAM

Software: Python 3.x, Jupyter Notebook or Google Colab, Anaconda (optional)

Python Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Joblib / Pickle

Viva Questions:

1. What is Ensemble learning?
2. What is the difference between Bagging and Boosting?
3. How does Random Forest reduce overfitting compared with a single Decision Tree?
4. Why do we use multiple trees in Random Forest?
5. What is the main idea behind Boosting?
6. What is the difference between AdaBoost and Gradient Boosting?
7. Why is cross-validation used in this experiment?
8. What does the standard deviation of cross-validation scores tell us?
9. What is feature importance in a tree-based model?
10. Which model would you choose for deployment: Decision Tree, Random Forest, AdaBoost, or Gradient Boosting?